4. Test Environment

4.1 Introduction

The test environment is the combination of hardware, software, tools, and system configurations used to execute the load testing activities. A properly configured test environment is essential for obtaining accurate and reliable performance test results.

For this project, the BlazeDemo Flight Booking Application was tested using Gatling on a local machine. The environment was configured to simulate virtual users performing flight booking operations and to collect performance metrics such as response time, throughput, request success rate, and error rate.

The details of the test environment used during the execution of the load tests are described below.

4.2 Hardware Configuration

The load tests were executed on a personal computer with the following hardware specifications:

Component	Configuration
Processor	Intel Core i5 / Intel Core i7 (As Available)
RAM	8 GB
Storage	256 GB / 512 GB SSD
System Type	64-bit Operating System
Network Connection	Broadband Internet Connection

The hardware resources were sufficient to execute Gatling simulations and generate performance reports without any resource limitations.

4.3 Software Configuration

The following software components were used to develop, execute, and analyze the load testing framework:

Software	Version
Java Development Kit (JDK)	JDK 21
Gatling	3.11.5
Apache Maven	Maven Build Tool
Eclipse IDE	Eclipse IDE for Java Developers
Google Chrome	Latest Stable Version
Operating System	Windows 10 / Windows 11

These software components were installed and configured before executing the load testing scenarios.

4.4 Operating System

The load testing activities were performed on the Microsoft Windows operating system.

Details:

- Operating System: Windows 10 / Windows 11
- Architecture: 64-bit
- User Environment: Local Development Machine

The operating system provided the runtime environment required to execute Java applications, Maven builds, and Gatling simulations.

4.5 Java Version

Java was used as the primary programming language for implementing the Gatling simulations.

Configuration:

- Java Version: JDK 21
- Compiler Version: Java 21
- Runtime Environment: Java Virtual Machine (JVM)

The Gatling simulation scripts were developed using the Gatling Java DSL and compiled using JDK 21.

Java was responsible for:

- Executing Gatling simulations
- Processing HTTP requests
- Managing virtual users
- Generating execution reports

4.6 Gatling Version

Gatling was used as the performance testing tool for this project.

Tool Details:

- Tool Name: Gatling
- Version: 3.11.5
- Framework Type: Open-Source Performance Testing Tool

Gatling was selected because it provides:

- High-performance load generation
- Java DSL support
- Detailed HTML reports
- Easy scenario implementation
- Efficient virtual user simulation

The tool was configured using Maven dependencies and executed through Java-based simulation classes.

4.7 IDE Used

The project was developed using Eclipse IDE.

IDE Details:

- IDE Name: Eclipse IDE for Java Developers
- Purpose: Development and Execution of Gatling Scripts

Eclipse was used for:

- Creating Java classes
- Managing Maven dependencies
- Writing Gatling simulation scripts
- Executing load tests
- Debugging execution issues

The IDE provided an integrated environment for developing and managing the entire load testing framework.

4.8 Browser Used

Google Chrome was used during the project for:

- Accessing the BlazeDemo application
- Capturing HTTP requests
- Inspecting network traffic
- Identifying request parameters
- Validating application workflows

Browser Details:

- Browser Name: Google Chrome
- Developer Tools: Chrome DevTools (F12)

Chrome Developer Tools were used to capture the HTTP requests associated with:

1. Search Flight Workflow
2. Choose Flight Workflow
3. Complete Booking Workflow

The captured request parameters were then implemented in the Gatling simulation scripts to create realistic user scenarios.

4.9 Environment Summary

The test environment consisted of a Windows-based system running Java 21, Gatling 3.11.5, Maven, Eclipse IDE, and Google Chrome. This environment provided all the necessary tools and resources required to design, execute, and analyze the load testing scenarios for the BlazeDemo Flight Booking Application.

The environment remained stable throughout the test execution process and successfully generated detailed performance reports for all implemented scenarios.